

(54) Title of the invention : A SYSTEM AND METHOD FOR BREAST CANCER DIAGNOSIS

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(57) Abstract :
ABSTRACT A SYSTEM AND METHOD FOR BREAST CANCER DIAGNOSIS The present disclosure discloses a system (100) for diagnosing breast cancer that utilizes topological data analysis to transform mammogram images into meaningful diagnostic insights. It includes a data preprocessing module (102) for image standardization and enhancement and a feature extraction module (104) to create histograms for topological analysis. The topological data analysis module (106) converts these histograms into Persistent Homology Diagrams (PHDs), representing topological features. An Earth Mover's Distance (EMD) matrix is generated by a similarity metric module (108) to compare PHDs. Representative PHDs are identified using a representative selection module (110), enabling accurate classification by the classification module (112). The system's performance is assessed through various metrics by a performance analysis module (114), and a web service module (116) provides an intuitive interface for users to upload images and receive diagnostic results. This approach enhances breast cancer detection by focusing on persistent topological features, offering improved precision and interpretability.

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